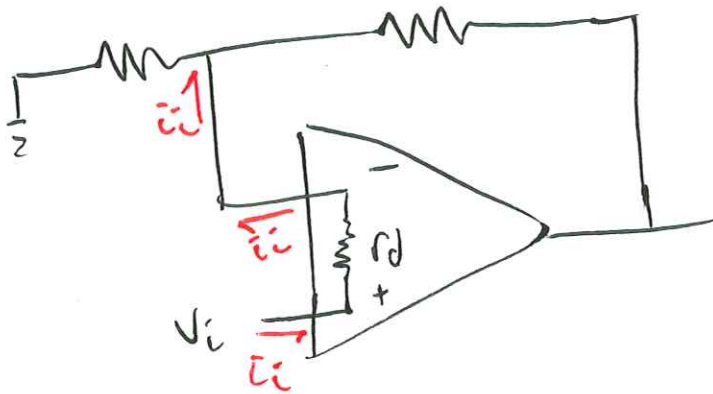


Op Amp Topologies → Consider where current is flowing
for this analysis, assume $r_i \neq \infty$, $r_o \neq 0$, $a \neq \infty$

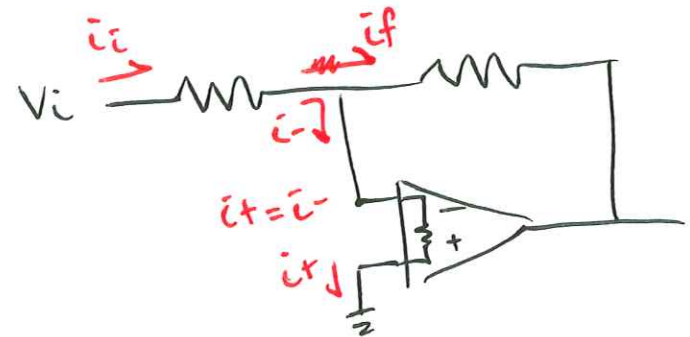
Inputs:

Series: Input current is NOT split by the feedback
" " " IS " "

Series:



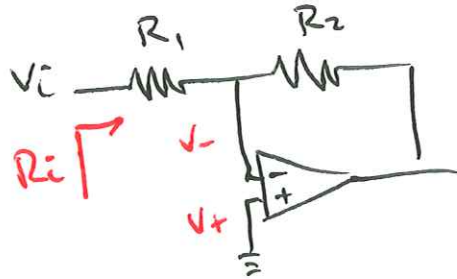
Shunt:



Significantly affects input resistance behavior

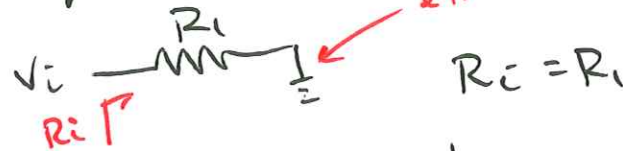
Input Resistance

Ex 1 (Shunt)



if $a \rightarrow \infty$, $V^- = V^+ = 0$

Effectively



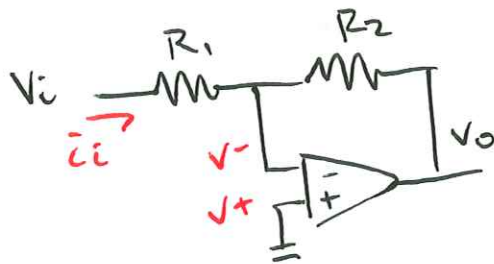
$$R_i = R_1$$

if $a \neq \infty$ but really large...

$V^+ \approx V^-$ but $\neq$

$$R_i \approx R_1$$

for this portion, assume $r_d = \infty$



$$V_d = a(V^+ - V^-) = -aV^-$$

$$V^- = (V_o - V_i) \frac{R_1}{R_1 + R_2} + V_i$$

$$V^- = -\frac{R_1}{R_1 + R_2} aV^- + V_i \left(1 - \frac{R_1}{R_1 + R_2}\right)$$

$$V^- = V_i \frac{1 - R_1/(R_1 + R_2)}{1 + aR_1/(R_1 + R_2)}$$

$$R_i = V_i / i_i$$

$$V_i - V_n = R_1 i_i$$

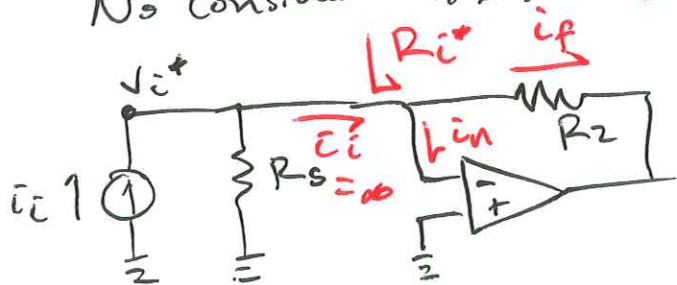
$$i_i = \frac{V_i - V_n}{R_1}$$

$$i_i = \frac{V_i}{R_1} \left[1 - \frac{1 - R_1/(R_1 + R_2)}{1 + aR_1/(R_1 + R_2)} \right]$$

$$R_i = R_1 / \left[1 - \frac{1 - R_1/(R_1 + R_2)}{1 + aR_1/(R_1 + R_2)} \right]$$

(still short)

No consider $r_o \neq \infty$ (eg. $1M\Omega$)



Norton equivalent source!
Thevenin: V_i^* R_s

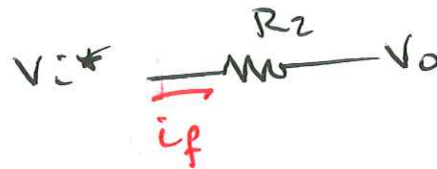
consider i_n vs i_f



$V_i^* = V^- \approx V^+$
(eg. $V^+ - V^- \approx 100\mu V$)

$V^+ - V^- = i_n r_o$

i_n in nA-pA range.
Exceedingly small!



$$V_i^* - V_o = i_f R_2$$

usually on order of mV-V

R_2 on order of $k\Omega - M\Omega$

i_f in mA range.

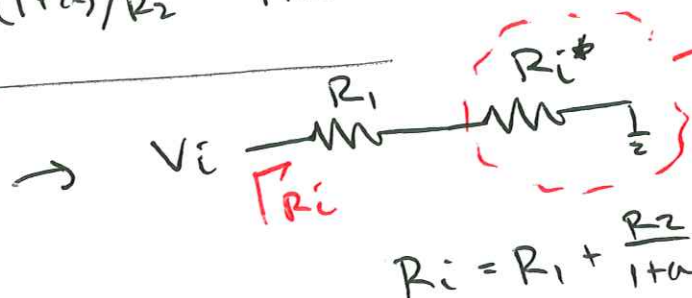
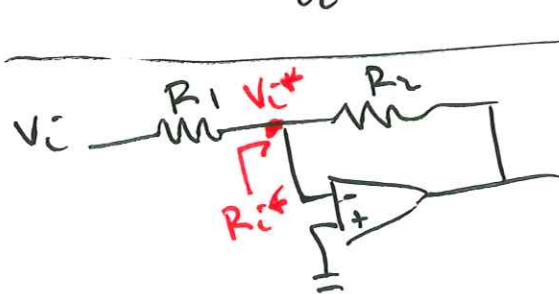
$i_f \gg i_n \rightarrow$ ignore r_o !

Analysis with $i_n = 0$, $r_o = \infty$

$$V_o = a(V^+ - V^-) = -aV_i^*$$

$$i_i = i_f = \frac{V_i^* - V_o}{R_2} = \frac{V_i^* (1+a)}{R_2}$$

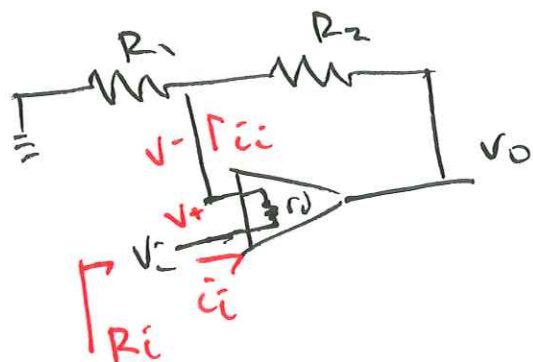
$$R_{i^*} = \frac{V_i^*}{i_i} = V_i^* / (V_i^* (1+a) / R_2) = \frac{R_2}{1+a}$$



representation of rest of circuit.

Series Input

$r_d \neq \infty$, ~~r_o~~ $R_2 \gg r_o \rightarrow$ assume $r_o = 0$
 $a \neq \infty$ (no significant output load)



$$V_d = V_i - V^- = V_i - bV_o = V_i - LV_i$$

$$V_i = (1+L)V_d$$



$$V_i - V^- = i_i r_d = V_d$$

$$R_i = \frac{V_i}{i_i} = r_d(1+L)$$

R_i effectively ∞ ! with reasonable L

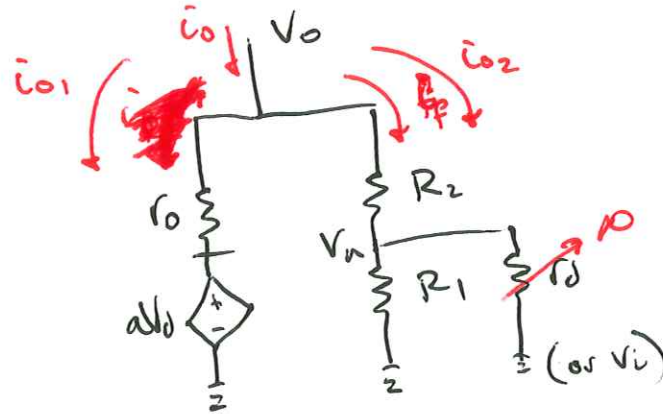
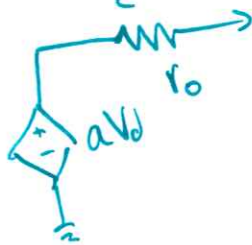
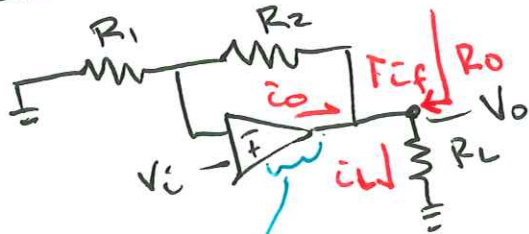
$\rightarrow$ assume $i_i \rightarrow 0$

Output Topologies:

Series: Output current is not split by feedback and load

shunt: " " " IS split " "

Shunt: (assume $r_d = \infty$), $r_o \neq 0$



$$\bar{i}_{o1} = \frac{aV_d}{r_o} = \frac{V_o - aV_d}{r_o}$$

$$V_d = V_o - V_i = \frac{R_1}{R_1 + R_2} V_o$$

$$\bar{i}_{o2} = \frac{V_o}{R_1 + R_2}$$

$$\bar{i}_o = \bar{i}_{o1} + \bar{i}_{o2} = \frac{V_o - aV_d}{r_o} + \frac{V_o}{R_1 + R_2}$$

$$\bar{i}_o = \frac{V_o (1 + a \frac{R_1}{R_1 + R_2})}{r_o} + \frac{V_o}{R_1 + R_2}$$

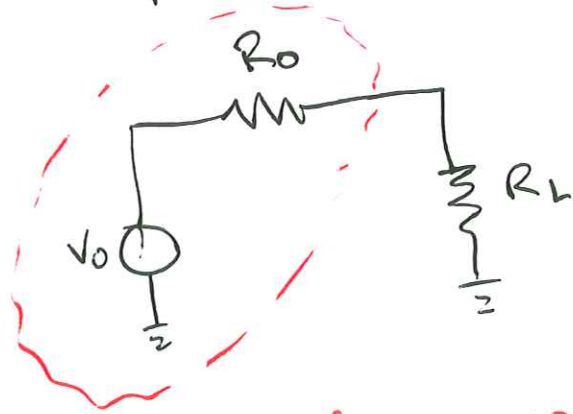
$$R_o = \frac{V_o}{\bar{i}_o} = r_o / [1 + L + \frac{r_o}{R_1 + R_2}]$$

usually $R_1, R_2 \gg r_o$, then $\frac{r_o}{R_1 + R_2} \ll 1$

then $R_o \approx r_o / (1 + L)$ with $L \rightarrow \infty$ $R_o \rightarrow 0$
ideal voltage source!

for output impedance analysis, assume all input sources are 0
 $\therefore V_i = 0$

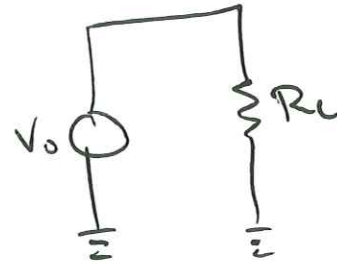
Output short Now modelled as:



Model of op-amp circuit

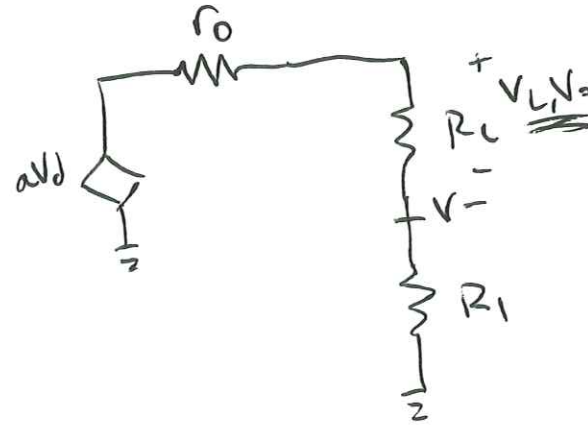
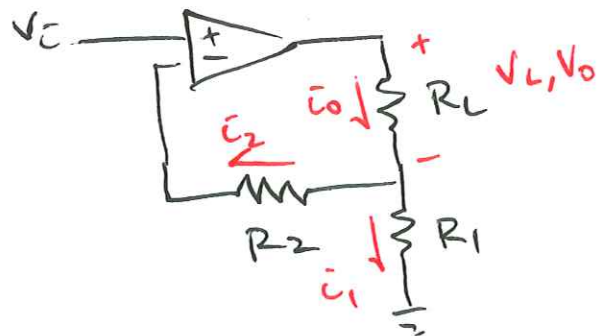
$$R_o \rightarrow 0$$

$\therefore$



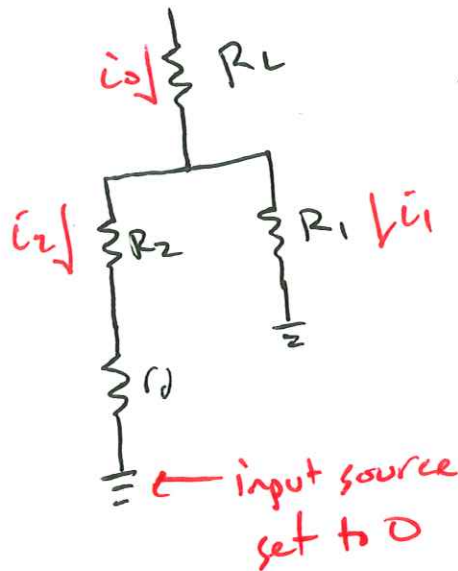
if $R_o \approx R_L$, then distorted voltage on R_L

Series Output



if $R_2 \ll r_d$,
then very small
voltage drop ~~across~~
across R_2

if $r_d \neq \infty$, then feedback:



contains R_L

$$aV_d = -R_1 i_o - r_o i_o + V_o$$

$$V_o = aV_d + R_1 i_o + r_o i_o$$

$$V_d = V^+ - V^- = R_1 i_o$$

if $R_1 \gg r_o$

$$R_o = \frac{V_o}{i_o} = R_1 (1 + a) \approx R_1 (1 + L)$$

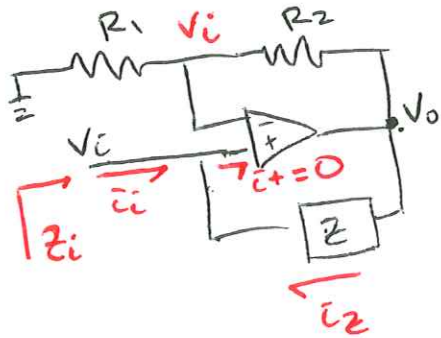
$L \rightarrow \infty, R_o \rightarrow \infty$ Ideal current source!

$$R_2 + r_d \gg R_1$$

$$\therefore i_1 \gg i_2$$

$$i_2 \rightarrow 0$$

Input Impedance Example.



$$\alpha = \infty, r_d = \infty, r_o = 0$$

$$V_i = V_o \frac{R_1}{R_1 + R_2} \Rightarrow V_o = V_i \left(1 + \frac{R_2}{R_1}\right) \leftarrow \text{from top feedback (-)}$$

$$(V_o - V_i) = Z i_z = -Z i_i$$

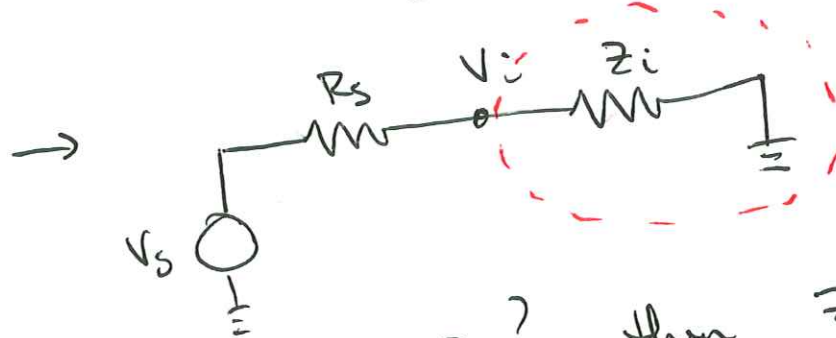
$\leftarrow$ from bottom feedback (+)

$$i_i = -i_z$$

$$i_i = \frac{V_o - V_i}{-Z} = \frac{V_i \left(1 + \frac{R_2}{R_1}\right) - V_i}{-Z} = -\frac{R_2 V_i}{R_1 Z}$$

Negative Impedance Converter (NIC)

$$Z_i = \frac{V_i}{i_i} = -\frac{R_1}{R_2} Z$$

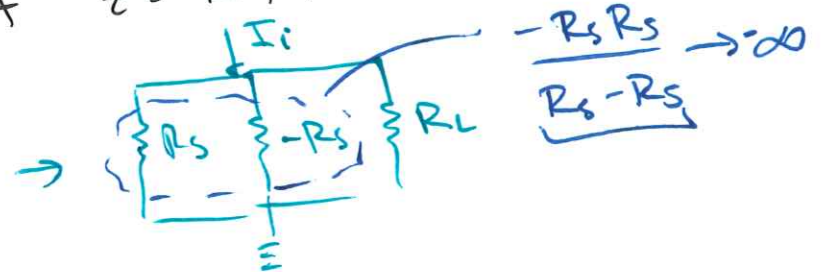
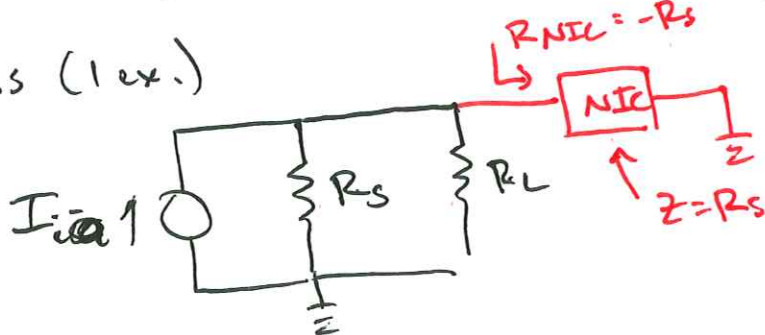


circuit representation.

what if $R_1 = R_2$? then $Z_i = -Z$

if $Z = R$, then $Z_i = -R$

usefulness (1 ex.)



$$\frac{-R_s R_s}{R_s - R_s} \rightarrow \infty$$